



# Preliminary analysis of NGT cluster usage and proposal

Felice Pantaleo (CERN)  
July 2026

# Method

## Data

Extracted pod lifecycle, requests, placement, cordons and per-GPU utilization from cluster monitoring, 30 days at 5 min resolution.

## Scope

1571 GPU/MIG requests from 114 users on the on-premise pools. STEAM Academy accounts and cloud T4 nodes excluded.

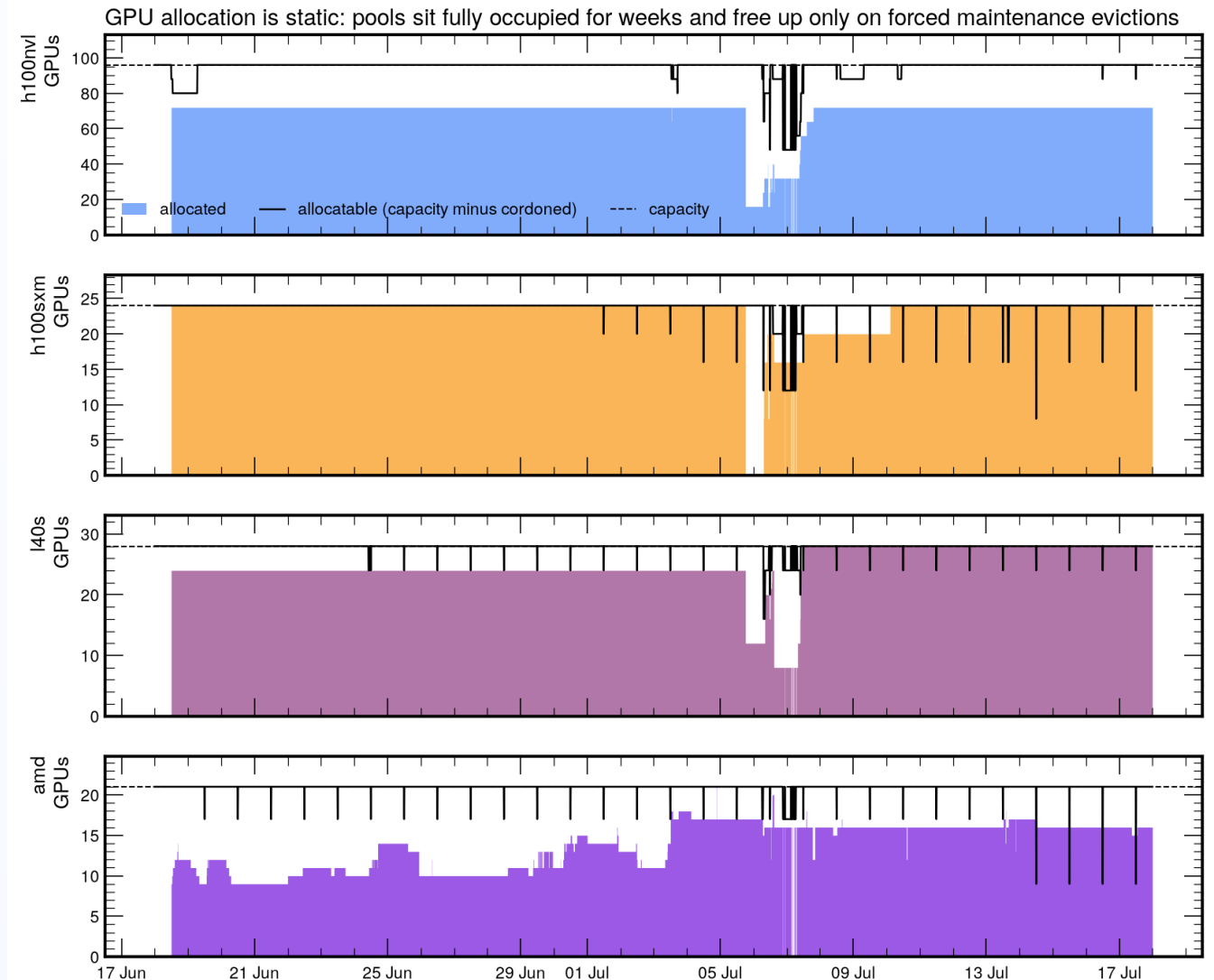
## Attribution

Users mapped to WP1 to WP3 via the project roster, group and manual classification. WP1 users possibly underestimated.

**A request = one pod asking for GPUs. Wait = creation to scheduling.**  
**Idle = GPU utilization below 5%.**

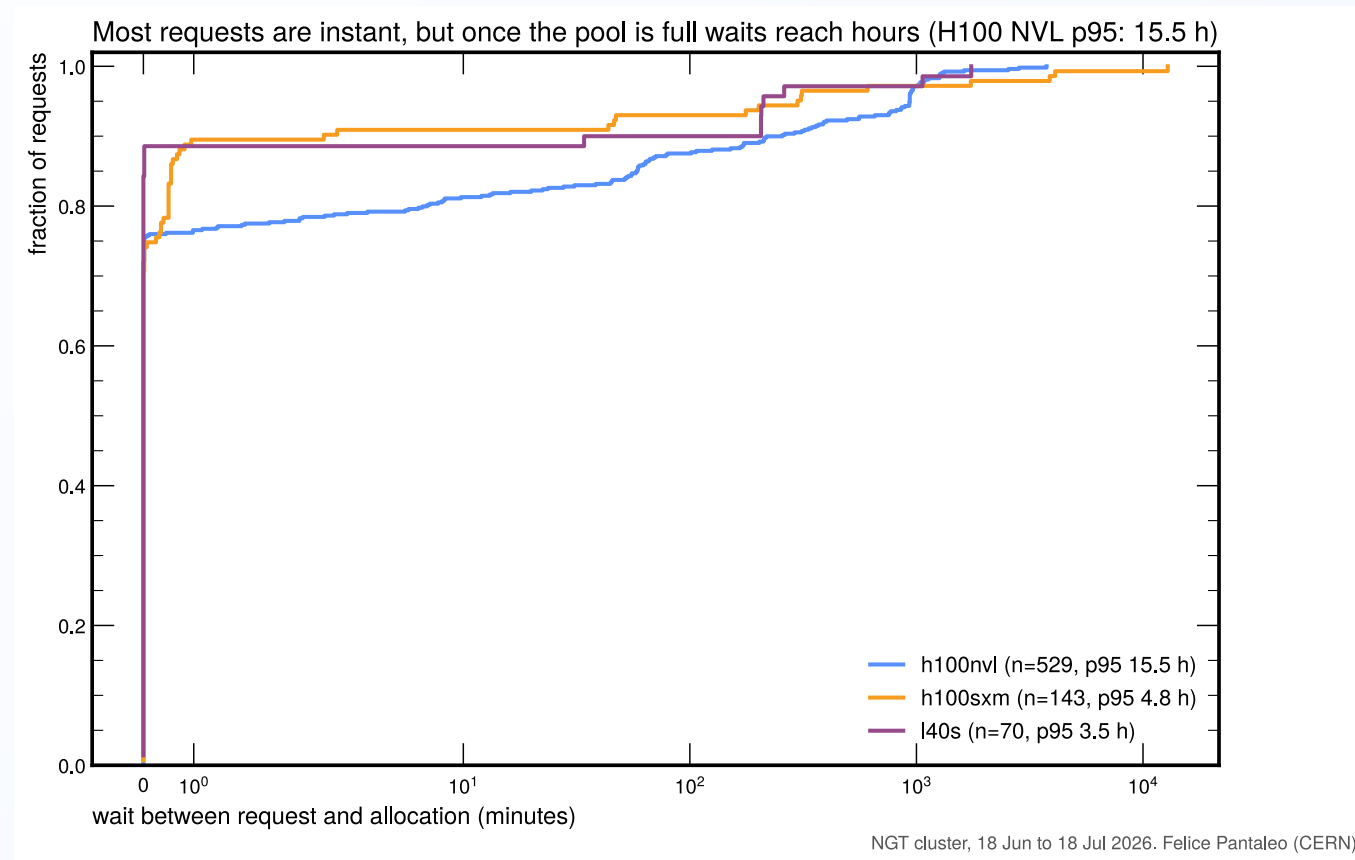
# Static allocation

- All pools at their effective ceiling around the clock
- No diurnal variation: allocation follows ownership, not workload
- Only the forced maintenance evictions of 6 to 8 July freed capacity



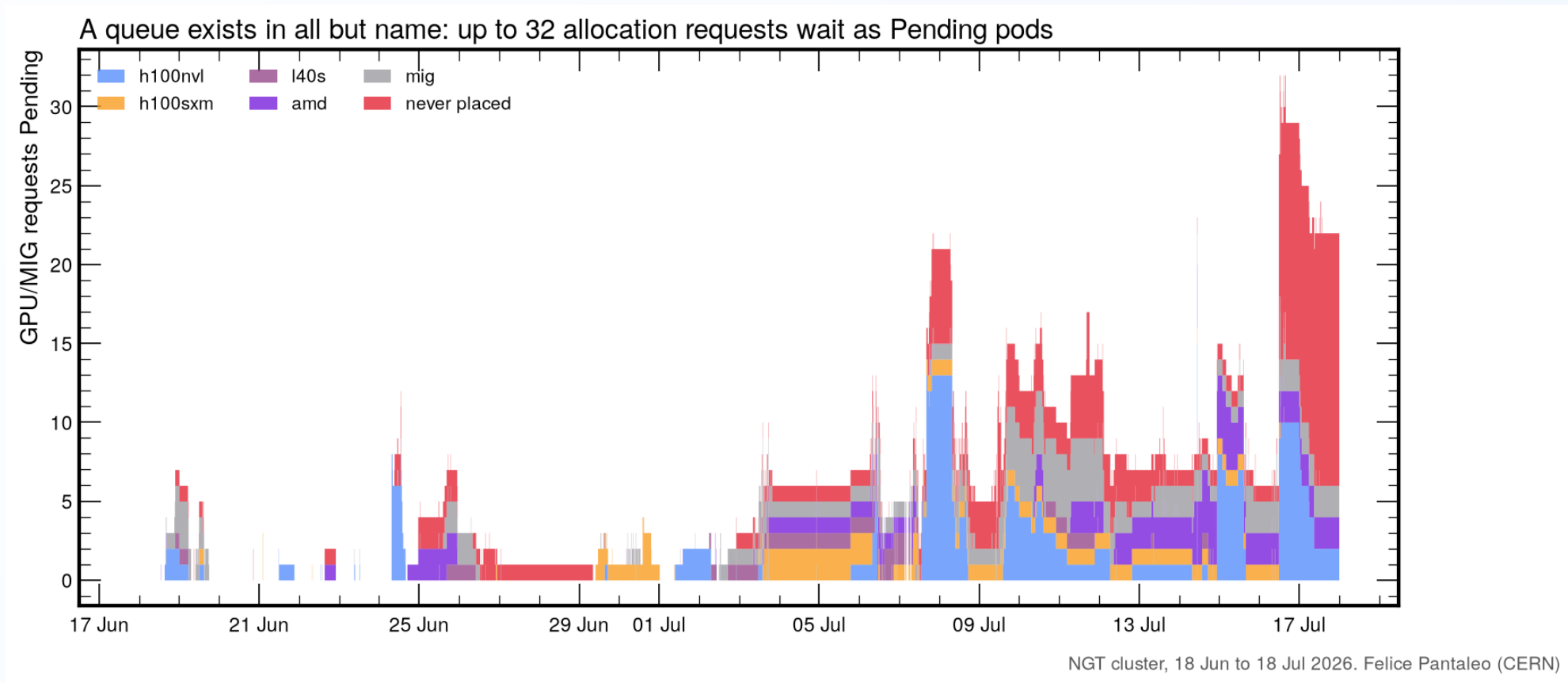
NGT cluster, 18 Jun to 18 Jul 2026. Felice Pantaleo (CERN)  
NVL: MIG-enabled GPUs are not visible to DCGM, so the full-GPU ceiling sits below the 96-GPU capacity line. AMD from user-pod accounting (no DCGM).

# Waiting times



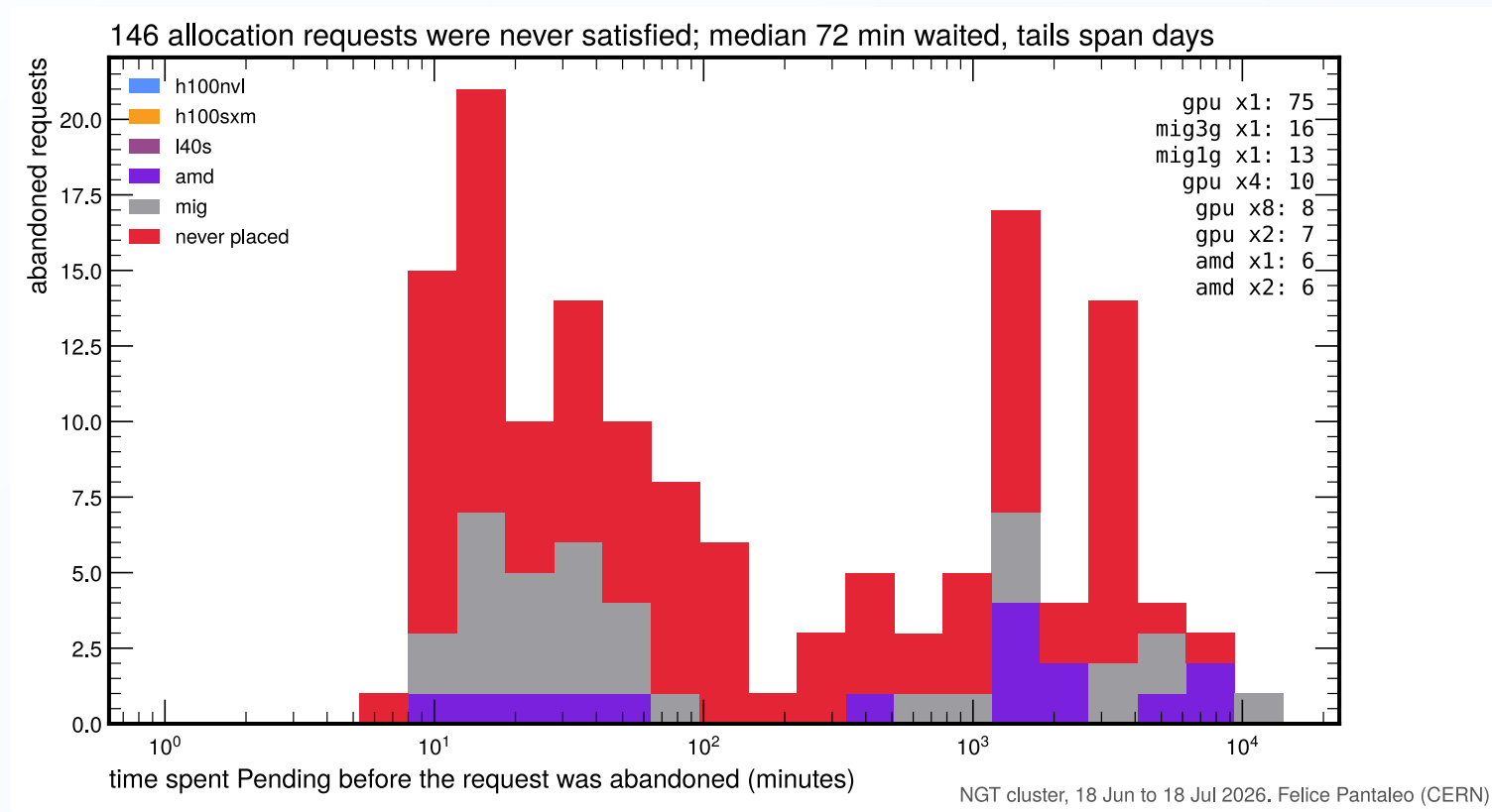
75 to 90% of requests are satisfied immediately; the H100 NVL p95 wait is 15.5 h.

# The Pending queue



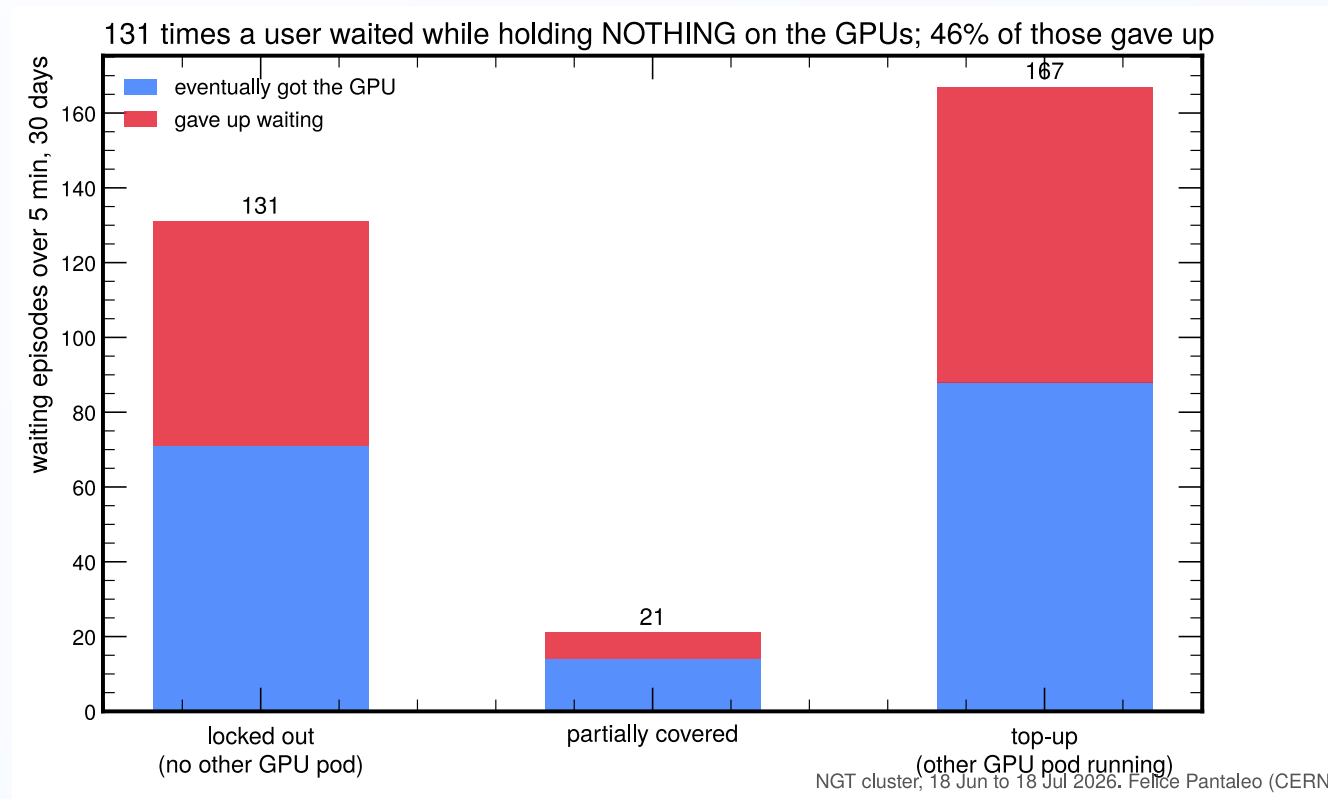
Peak backlog: 32 simultaneous Pending requests, stacked by target pool.

# Unsatisfied requests



146 requests (9%) were never satisfied; their owners gave up after a median 72 minutes of waiting.

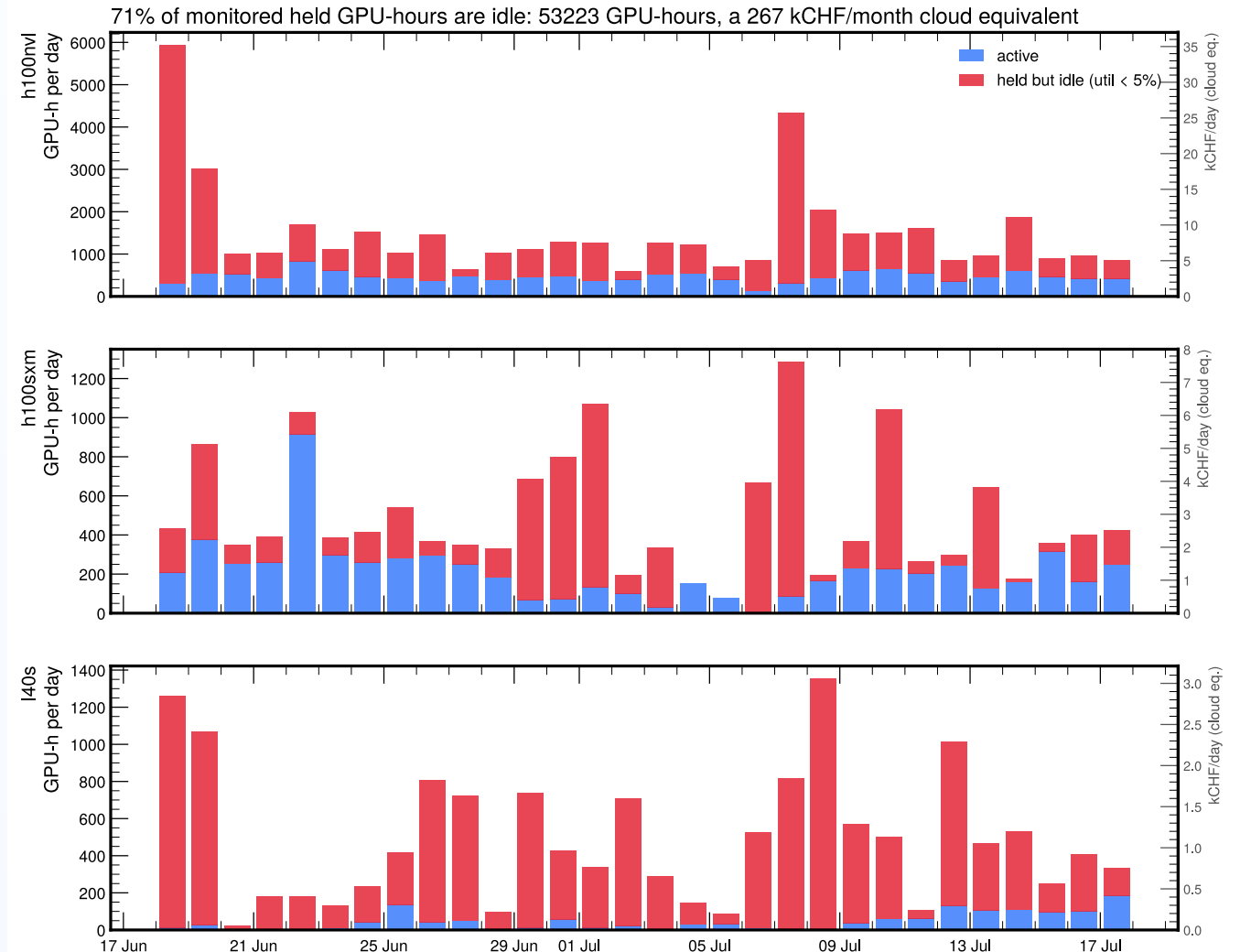
# Lockouts



131 times last month a new user not has been locked out at their first request.  
Over half of all queue pressure is top-up demand from users already running.

# Idle held GPU-hours

- 53 000 GPU-hours parked in 30 days
- 267 kCHF/month cloud equivalent
- Right axes: kCHF/day per pool
- Continuous idle baseline on NVL and L40S; bursty parked episodes on SXM

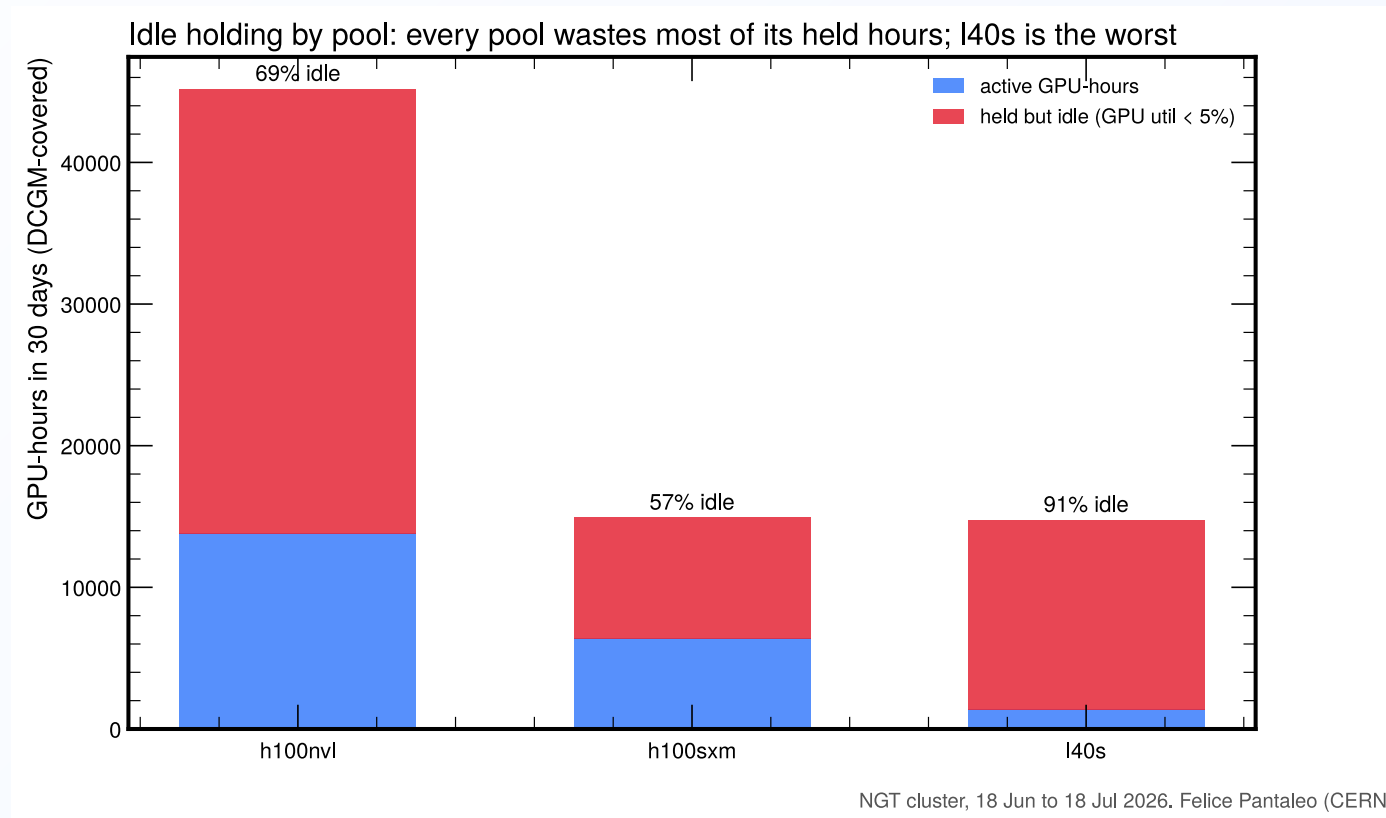


DCGM utilization available for 500 full-GPU allocations; MIG slices and AMD not covered. Cloud rates as in the cost plot.

NGT cluster, 18 Jun to 18 Jul 2026. Felice Pantaleo (CERN)

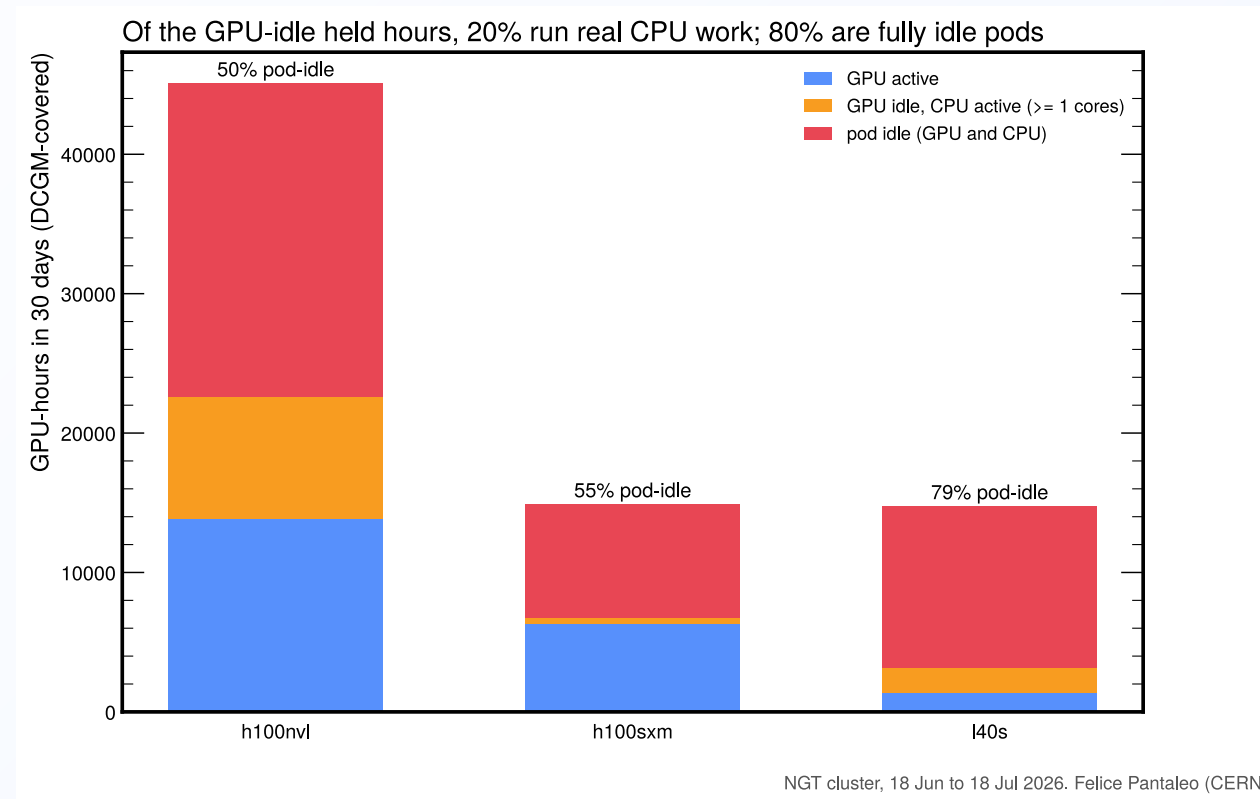


# Idle share by pool



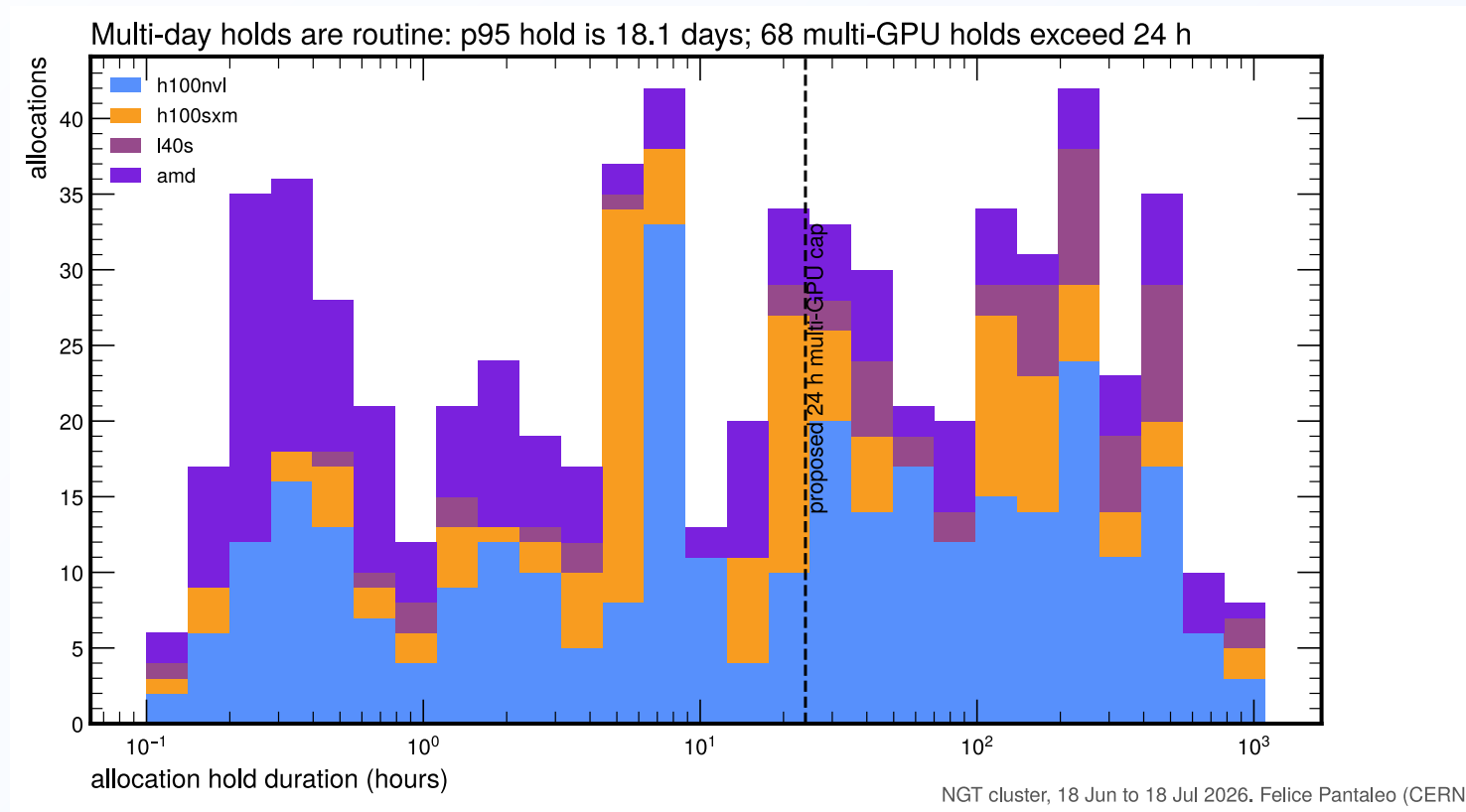
The L40S pool is idle for 91% of its held hours: mainly explained by spiky heterogeneous algorithms development, benchmarking, and fear of losing the pod/tmux session.

# GPU-idle vs pod-idle



Only 20% of GPU-idle hours run real CPU work ( $\geq 1$  core); 80% are fully idle pods. Robust across 0.5 to 2 core thresholds.

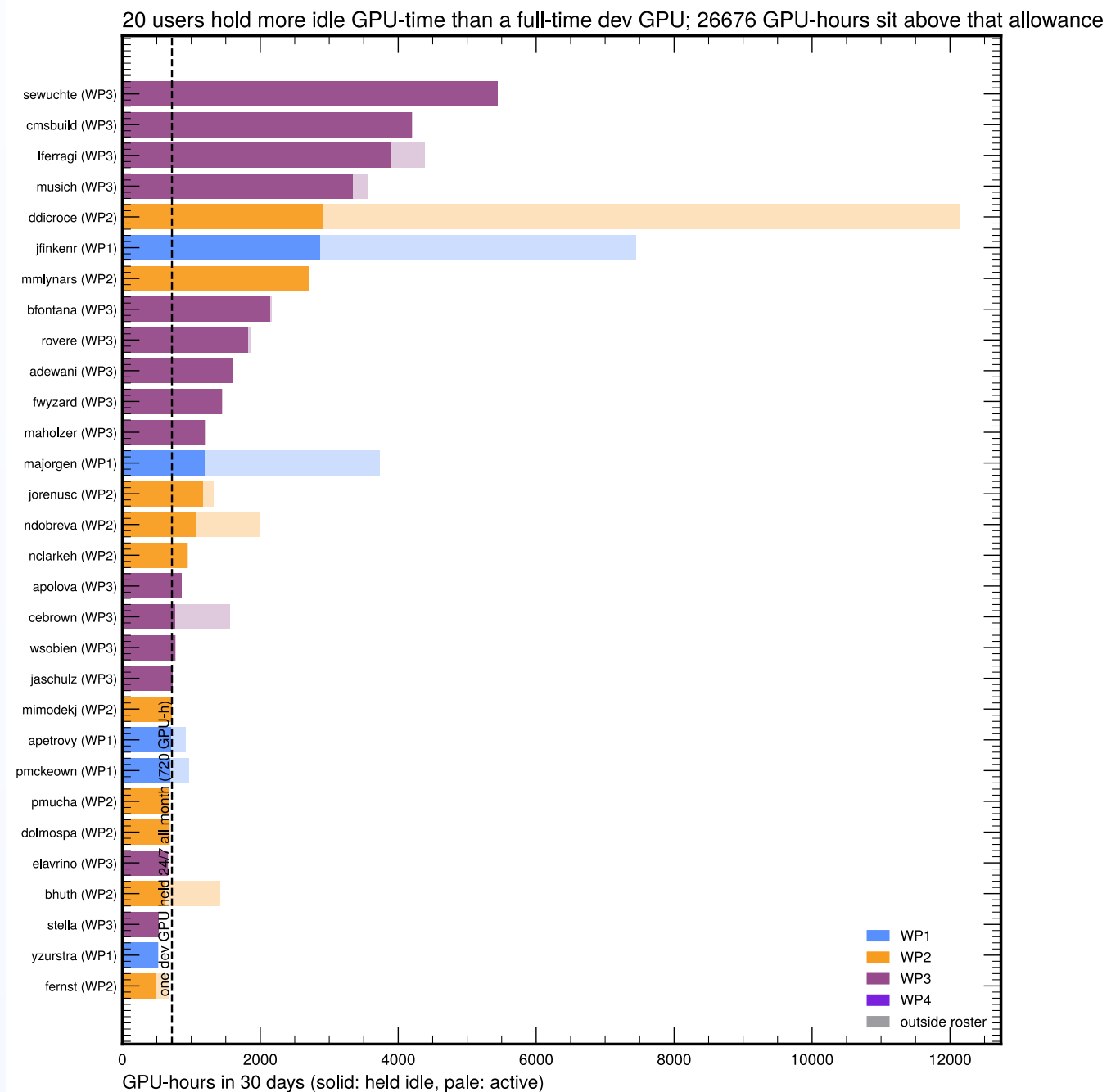
# Hold durations



The p95 hold is 16.7 days; 68 multi-GPU holds exceed 24 hours.

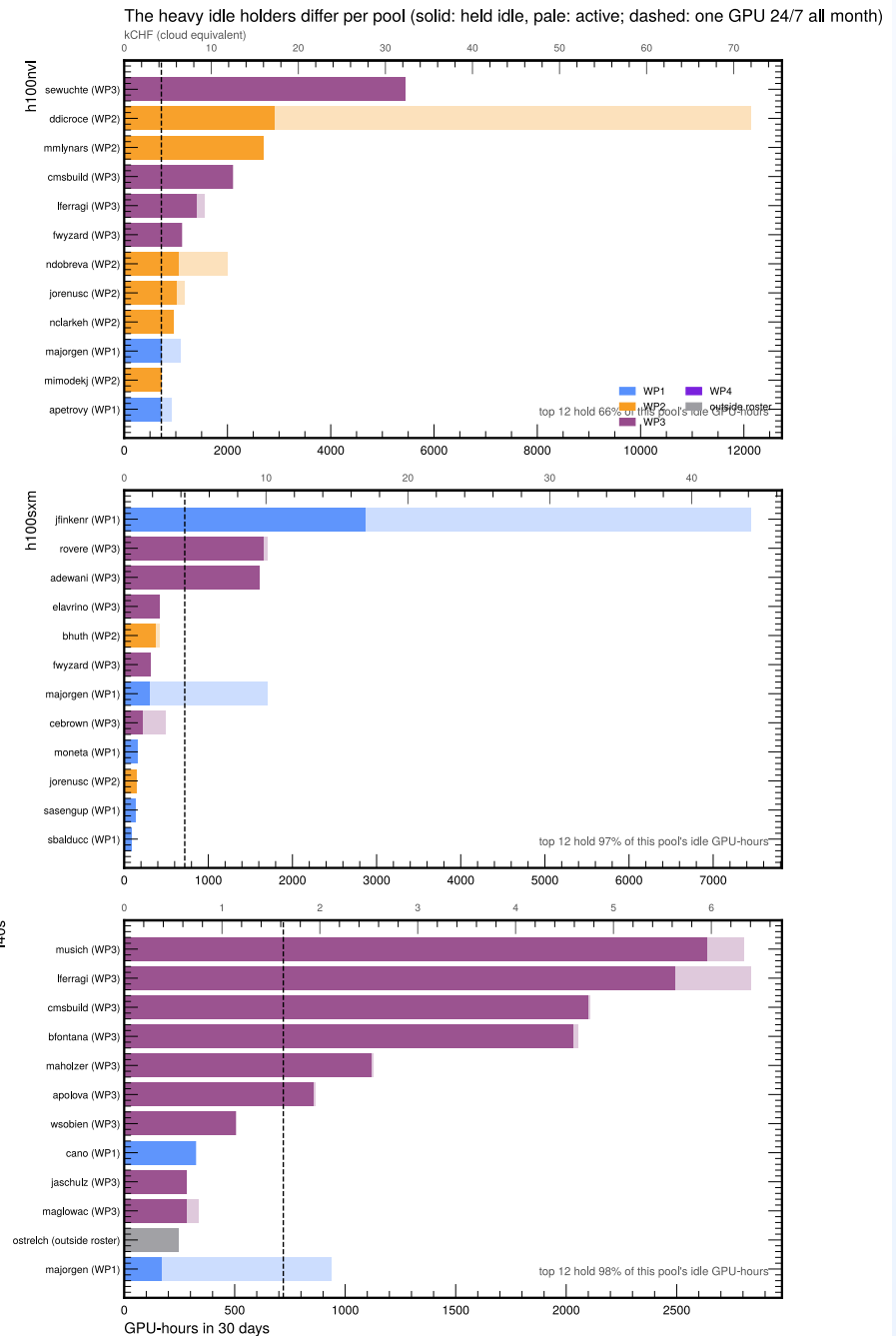
# Heavy idle holders

- Solid = held idle, pale = active
- Dashed line: one dev GPU holding 24/7 all month (720 GPU-h).
- **20 users** exceed it
- **26 700 GPU-hours** sit above it
- The consumption and idleness rankings select different users: the top consumer is almost entirely active

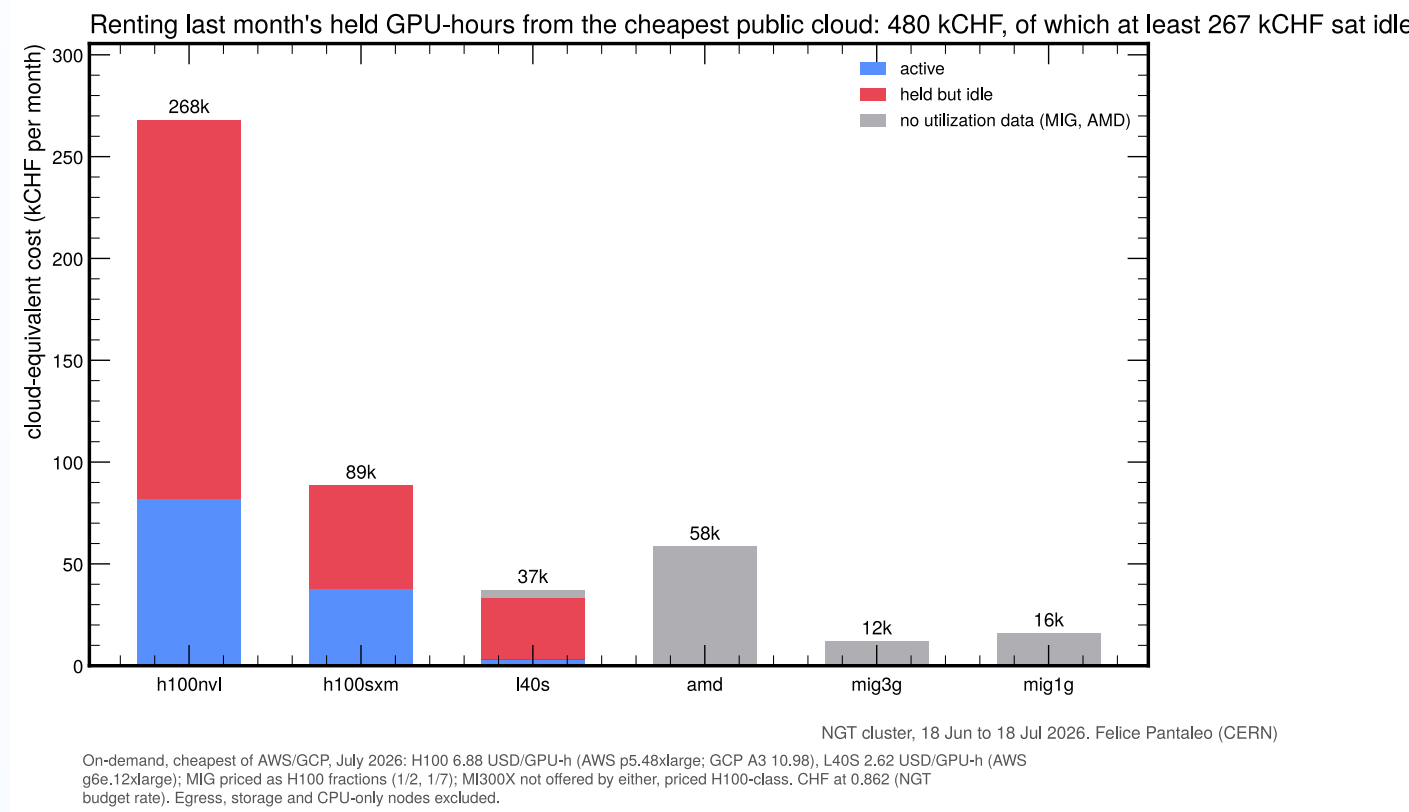


# Idle holders by pool

- H100-NVL: broad behavior, top 12 hold 66% of idle
- H100-SXM: three users own the idle hours. Do they all need fast inter-GPU communication?
- L40S: 12 users own 98% of the idle hours
- The one-session rule and batch-only multi-GPU bound all of this by construction

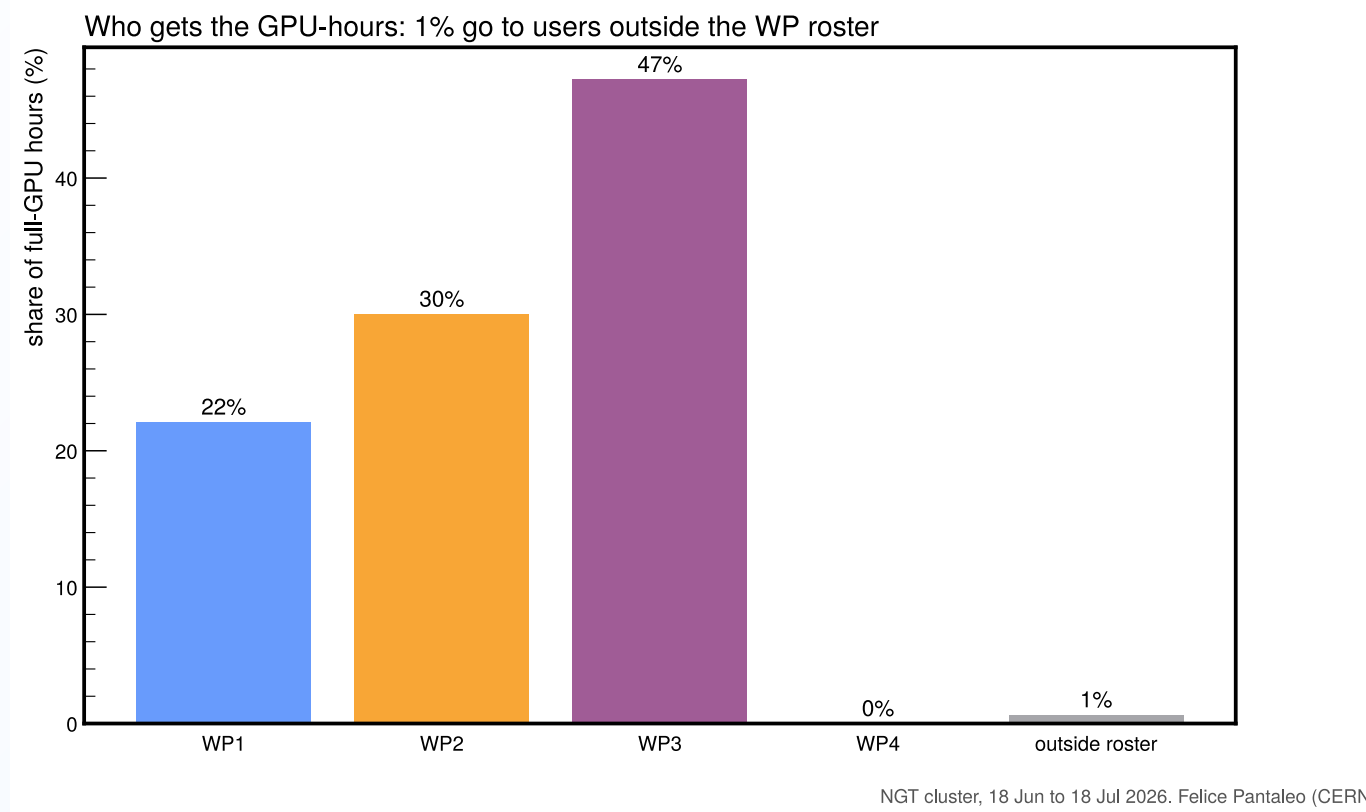


# Cloud-equivalent cost



Rental equivalent: 480 kCHF/month, of which at least 267 kCHF idle (3.2 MCHF/year). On-demand rates, July 2026 (AWS).

# GPU-hours by WP



# Proposal for a new policy

Simulated the same 30 days, same requests, replayed through the proposed policy:

|                                       | Today (FCFS) | Proposed      |
|---------------------------------------|--------------|---------------|
| wait p95                              | 557 min      | <b>0 min</b>  |
| requests satisfied                    | 96.7%        | <b>99.4%</b>  |
| dev sessions within 15 min            | 91%          | <b>98%</b>    |
| NVL idle-held GPU-hours               | 35 800       | <b>13 600</b> |
| running work terminated by the system | none         | <b>none</b>   |

**Policy:** one interactive session per member (a new session supersedes the previous one); multi-GPU beyond a 96 GPU-h/month interactive allowance runs as batch behind a WP fair-share priority queue.

**Fixed-behavior replay, identical requests for every policy; the FCFS baseline reproduces observed waits (median exact, p95 within factor 2).**



# Proposal to the PMC

## One interactive session

One single-GPU session per member, guaranteed and always available; opening a new one replaces the old.

## Multi-GPU is batch

Submitted, executed, exits at completion, behind a priority queue with WP fair share. A 96 GPU-h/month interactive allowance covers debugging. Amount to be tuned.

## Account and adjust

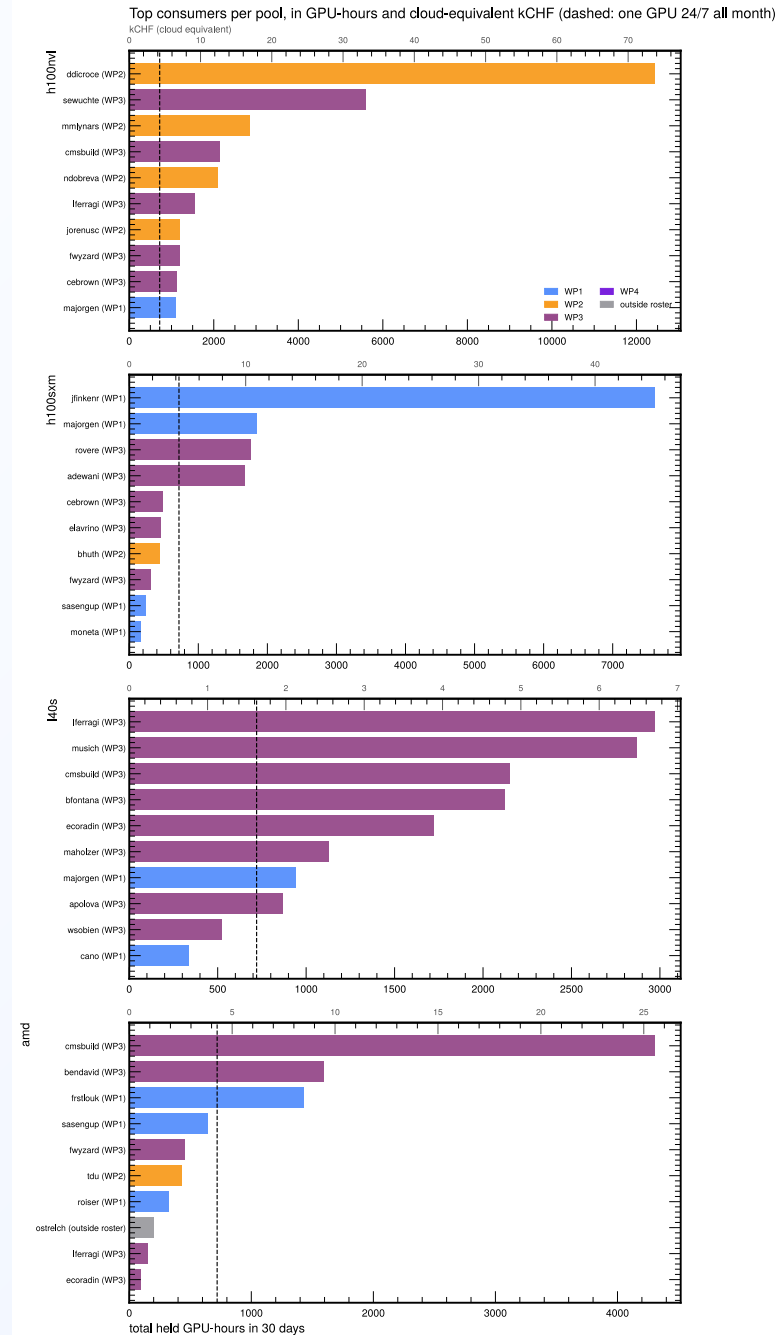
GPU time charged to WPs with model factors. All parameters simulated and tunable on the real trace.

# Backup



# Backup: top consumers per pool

- Secondary axis: cloud-equivalent kCHF, cheapest of AWS/GCP, July 2026
- Dashed: one GPU 24/7 all month
- 31 users held more than one GPU-month



# Backup: pools and cordons

